Stack

Data Structures and its Applications

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Quiz 1: Stack

A stack is a data structure which does not support the following function

a) push

b) pop

c) Is_empty

d) indexing

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Quiz 2: Stack

If a string is of the form wcw^R, where w is a string of characters not having c and w^R is the reverse of W, then it is a palindrome. To check for a such a palindrome, the size of the stack should be at least

a) The length of w

b) The length of wcw^R

c) The length of wc

d) None of these

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Quiz 3: Stack

To check whether a given string is of the form $a^n b^{2n}$, some number of a followed by twice the number of b, each time we get a

a) push a symbol into the stack

b) Push 2 symbols into the stack

c) Pop a symbol from the stack

d) None of these

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Quiz 4: Stack

In an array implementation of the stack, push should

a) Increment top and then use that the index

b) Use top as the index and then increase top

c) Use top and should not change it

d) None of these

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Quiz 4: Stack

In an array implementation of the stack, push should

a) Increment top and then use that the index

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Quiz 5: Stack

In an array implementation of the stack, if is_full returns true

a) The program should be aborted

b) The top element should be popped off

c) The stack should be reset

d) The user should decide what is the best action

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Quiz 5: Stack

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Stack : Abstract Data Type

stack

- Is a data structure
- Has zero or more components
- Can operate at only one end – normally called the stack top
- Supports LIFO : Last In First Out
- What goes in last, comes out first
- Does not support the concept of traversal

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Stack : Interface

stack

- Create
- Destroy
- Push : add an element at the top
- Pop : remove an element at the top
- Peek : check the top element without removing it
- Is_empty : check whether the stack is empty
- Is_full : check whether the stack is full

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Stack : Implementation

stack

- using
    - A linked list
    - An Array

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THANK YOU

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